Another algorithm might change the data from a *bitmap* to a *vector mapped* format. This involves creating a list of the ridges contained in the image and the pixels contained in each ridge. In the vector mapped form, each ridge in the fingerprint has an individual identity, as opposed to an image composed of many unrelated pixels. This can be accomplished by looping through the image looking for the endpoints of each line, the pixels that have only one black neighbor. Starting from the endpoint, each line is traced from pixel to connecting pixel. After the opposite end of the line is reached, all the traced pixels are declared to be a single *object*, and treated accordingly in future algorithms.

Computed Tomography

A basic problem in imaging with x-rays (or other penetrating radiation) is that a *two-dimensional image* is obtained of a *three-dimensional object*. This means that structures can *overlap* in the final image, even though they are completely separate in the object. This is particularly troublesome in medical diagnosis where there are many anatomic structures that can interfere with what the physician is trying to see. During the 1930's, this problem was attacked by moving the x-ray source and detector in a coordinated motion during image formation. From the geometry of this motion, a single *plane* within the patient remains in focus, while structures outside this plane become blurred. This is analogous to a camera being focused on an object at 5 feet, while objects at a distance of 1 and 50 feet are blurry. These related techniques based on motion blurring are now collectively called **classical tomography.** The word *tomography* means *"a picture of a plane."*

In spite of being well developed for more than 50 years, classical tomography is rarely used. This is because it has a significant limitation: the interfering objects are not *removed* from the image, only *blurred*. The resulting image quality is usually too poor to be of practical use. The long sought solution was a system that could create an image representing a 2D slice through a 3D object with *no* interference from other structures in the 3D object.

This problem was solved in the early 1970s with the introduction of a technique called **computed tomography (CT)**. CT revolutionized the medical x-ray field with its unprecedented ability to visualize the anatomic structure of the body. Figure 25-13 shows a typical medical CT image. Computed tomography was originally introduced to the marketplace under the names *Computed Axial Tomography* and *CAT scanner*. These terms are now frowned upon in the medical field, although you hear them used frequently by the general public.

Figure 25-14 illustrates a simple geometry for acquiring a CT slice through the center of the head. A narrow pencil beam of x-rays is passed from the x-ray source to the x-ray detector. This means that the measured value at the detector is related to the total amount of material placed *anywhere*